

Comparative Evaluation of Constrained and Unconstrained Deep Reinforcement Learning for Safe Robotic Grasping on a UR5e Manipulator in Simulation

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Introduction:

UR5: A collaborative arm that shares space with people

- 6-DOF, 5 kg payload, 850 mm reach
- Max joint velocity 180 deg/s (π rad/s)
- with $\pm 360^\circ$ working limit
- Works next to people, not behind a fence

We use Reinforcement Learning (RL) to train robots



Fig 1: A human operator working side by side with a UR5 manipulator

Background:

❑ Reward shapes Policy but Constrains add safety with consistency

- A learned policy can't be inspected like a hand-coded controller — its behaviour is only known statistically.
 - Safety on this class of arm is already an active industrial requirement, not a hypothetical
 - Validated only by trial and error, after the fact
 - This thesis tests the alternative: pull safety out of the reward
- Let's not just train a policy based on reward and penalty - **start adding constraints**

Literature Review:

Positioning Against the Closest Work

Property	Khan et al. [8]	Shen et al. [9]	This thesis
Manipulator	Franka Panda	4-DOF planar	UR5e
Safety expressed as	Constraint	Reward term	Constraint
Safety quantity	Collision cost	Manipulability	Manipulability + joint limit + collision
Algorithm	PPO-Lagrangian	SAC (unconstrained)	PPO-Lagrangian
Control arm	✗	✗	✓
Seed dispersion reported	✗	✓ (5 seeds)	✓ (5 seeds)
Cost Limit (d)	✗ (d=25 only)	n/a	✓ (d=25 and d=15)

Research Gap:

Three gaps in the closest published work and the design response

Gap	What the literature does	This thesis's response
No control arm	Trains cPPO vs. PPO, reports the raw difference	measures the implementation term directly
No dispersion	Mostly use only 1 seed	5 seeds for training, 3 for evaluation
One budget only	Fixes cost limit $(d) = 25$ only	Used cost limit $d = 25$ & 15 effect read against how hard the constraint

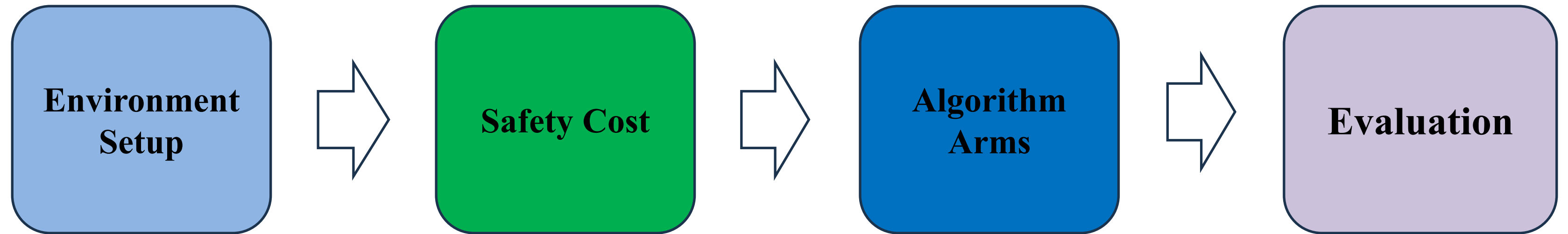
3 concrete gaps in the literature and our response to each: no control arm -- we add one; no dispersion -- we report five-seed spreads; one budget -- we test two.

Objectives:

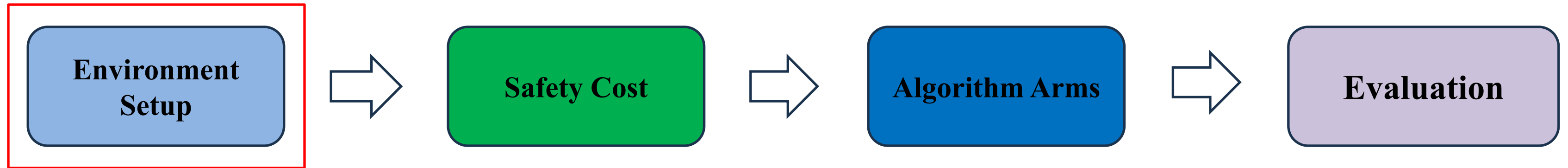
The main objectives of this thesis are to,

- Implement both the constrained and unconstrained agents
- Isolate the constraint as a cost critic with λ as a Lagrangian multiplier
- Use different seeds both for training and evaluation both

Methodology:

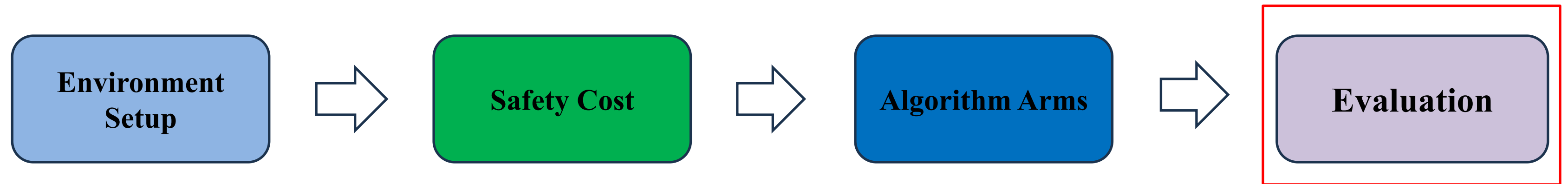


Methodology:



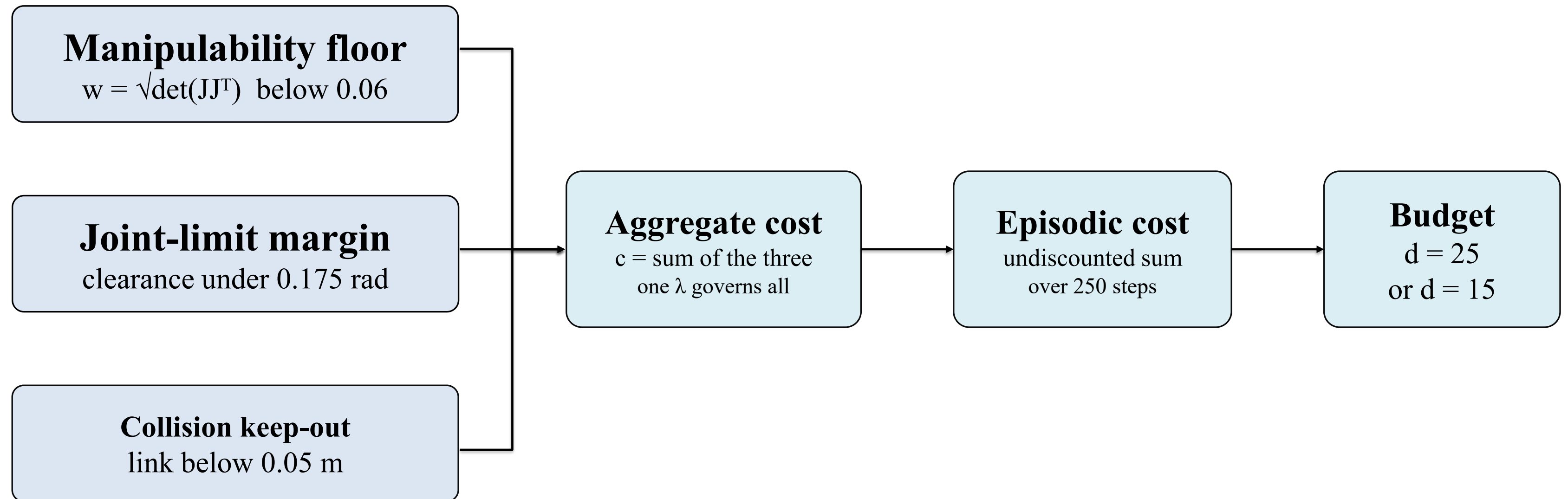
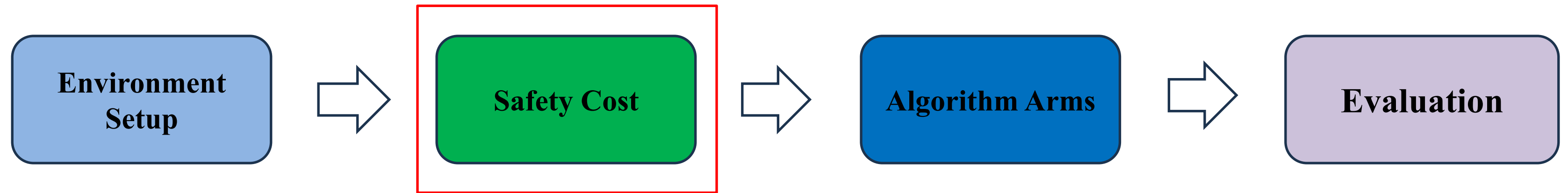
- NVIDIA IsaacSim 5.0.0
- IsaacLab 2.3.1
- rsl_rl framework
- weld environment
- **4096** parallel robots for training
- **1500** iterations
- UR5e + Robotiq 2F-85 gripper

Methodology:

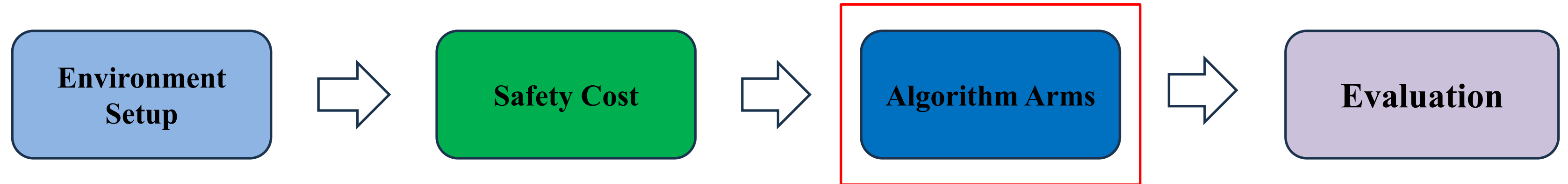


- **256** parallel robots
- 1000 episodes per arm,
total **45,000 episodes**
- 3 different seeds
- Collision keep-out -- **5** cm above table
- Joint-limit margin -- **0.175** rad (10°)
- Singularity floor $w = \sqrt{\det(JJ^T)}$ --
 $w < 0.06$

Methodology:



Methodology:



- Proximal Policy Optimization (PPO)
- PPO (baseline)
- Constraint Proximal Policy Optimization (cPPO)
- Budget Limit $d = 25$ & 15

Methodology:

The 3 arms, and what each one isolates

activate the budget

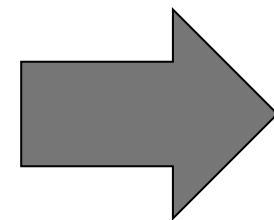
tighten the budget

PPO (baseline)

cost critic present

λ **ceiling** = 0

the unconstrained reference

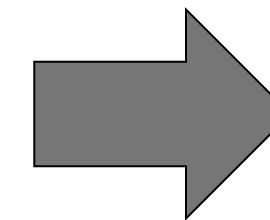


cPPO

cost critic free

budget d = 25

constraint active



cPPO tightened

cost critic free

budget d = 15

binds on every seed

isolates the constraint effect — the claim of this thesis

isolates budget sensitivity — does it scale?

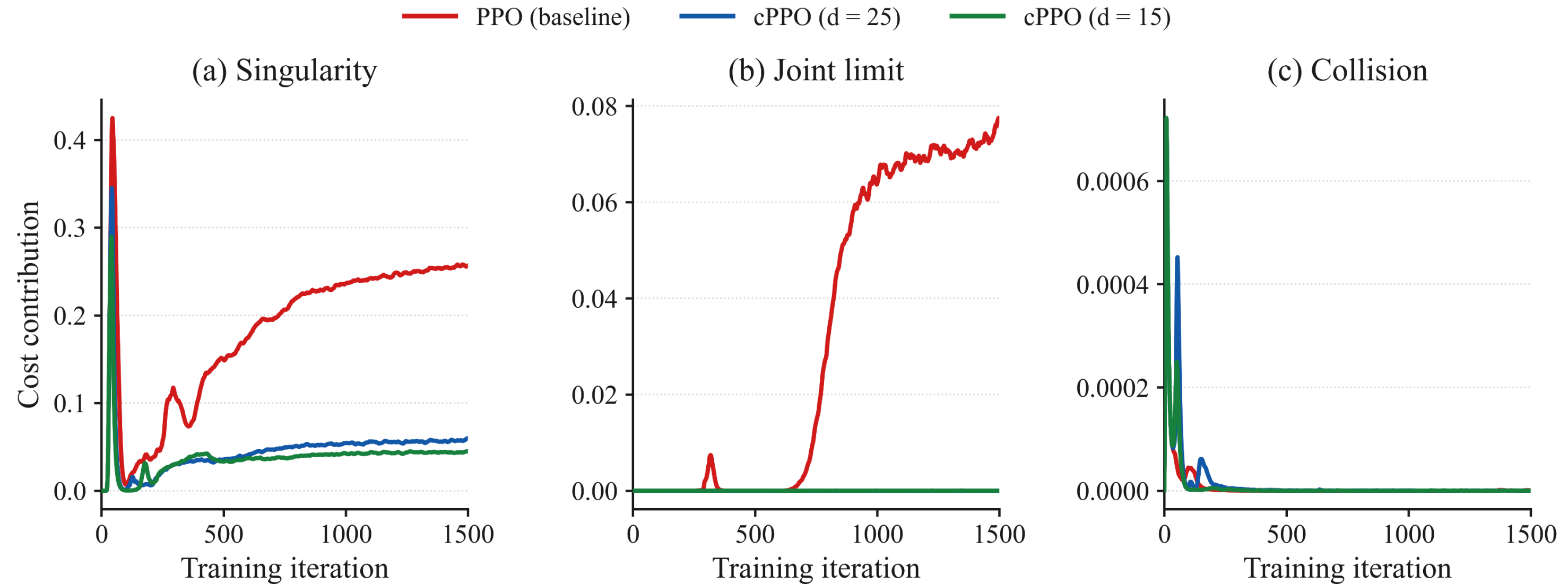
5 seeds per arm (1, 3, 4, 52, 54) → 15 trained policies.

Every quantity is reported with its spread across seeds.

RESULT:

Mean Reward: Lifting-object reward:

• PPO(baseline) :	133.06	9.467
• cPPO(d=25) :	132.18	9.437
• cPPO(d=15) :	134.09	9.441

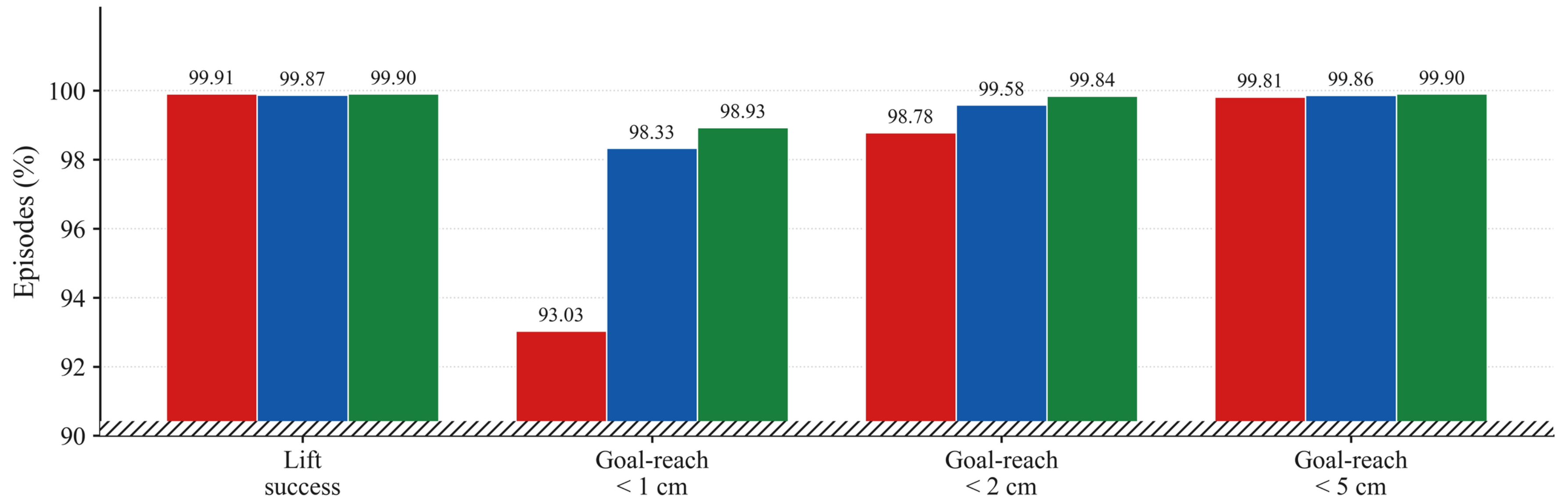


RESULT:

The constraint costs nothing measurable
Constraining the policy did not cost task performance at either budget.

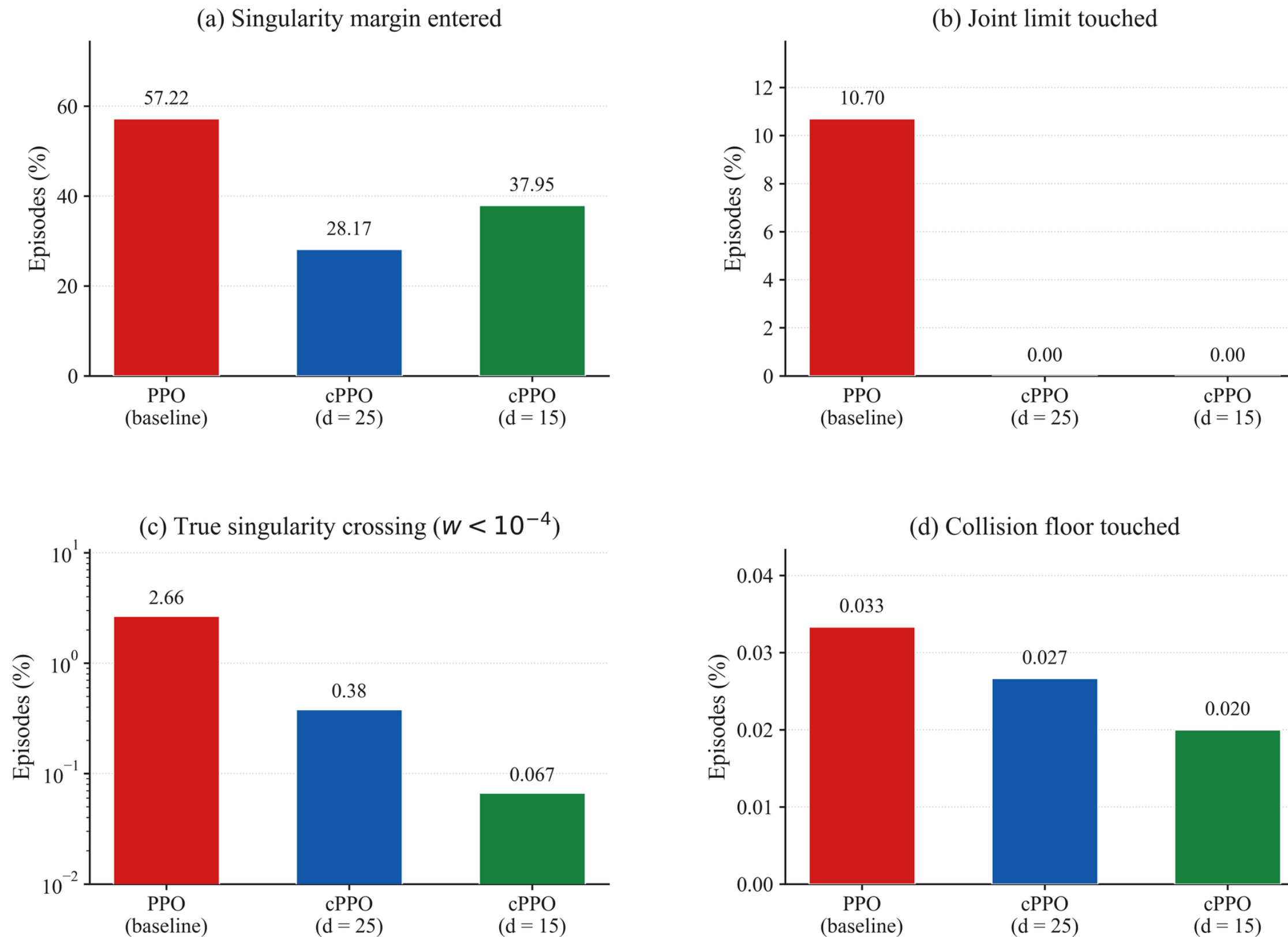
— PPO (baseline) — cPPO (d = 25) — cPPO (d = 15)

(a) Success rate (axis zoomed; 0--90 % suppressed)



RESULT:

Safety: fewer true singularities, zero joint-limit contact



Conclusion:

A stated budget changes what a policy does — mostly by making it predictable

- Constraining the policy reduces unsafe behaviour
- At no measurable task cost.
- An unanticipated effect is on consistency

Future Work:

- IBVS systems
- PID Lagrangian
- Sim-to-real application
- More Seeds implementations

Key Resources:

- [1] Altman, E. (1999). Constrained Markov Decision Processes. Chapman & Hall/CRC.
- [2] Schulman, J. et al. (2017). Proximal Policy Optimization Algorithms. arXiv:1707.06347.
- [3] Achiam, J. et al. (2017). Constrained Policy Optimization. Proc. ICML, PMLR 70.
- [4] Ray, A., Achiam, J., & Amodei, D. (2019). Benchmarking Safe Exploration in Deep RL. OpenAI.
- [5] Stooke, A., Achiam, J., & Abbeel, P. (2020). Responsive Safety in RL by PID Lagrangian Methods. Proc. ICML, PMLR 119.
- [6] Yoshikawa, T. (1985). Manipulability of Robotic Mechanisms. Int. J. Robotics Research, 4(2).
- [7] Henderson, P. et al. (2018). Deep Reinforcement Learning That Matters. Proc. AAAI 32(1).
- [8] Khan, F. et al. (2026). RL for Precision Grasping and Safety-Critical Coordination in a Robotic Arm. Intelligent Service Robotics, 19(16).
- [9] Shen, Y. et al. (2022). RL-Based Reactive Obstacle Avoidance for Redundant Manipulators. Entropy, 24(2).

THANK YOU

Appendix:

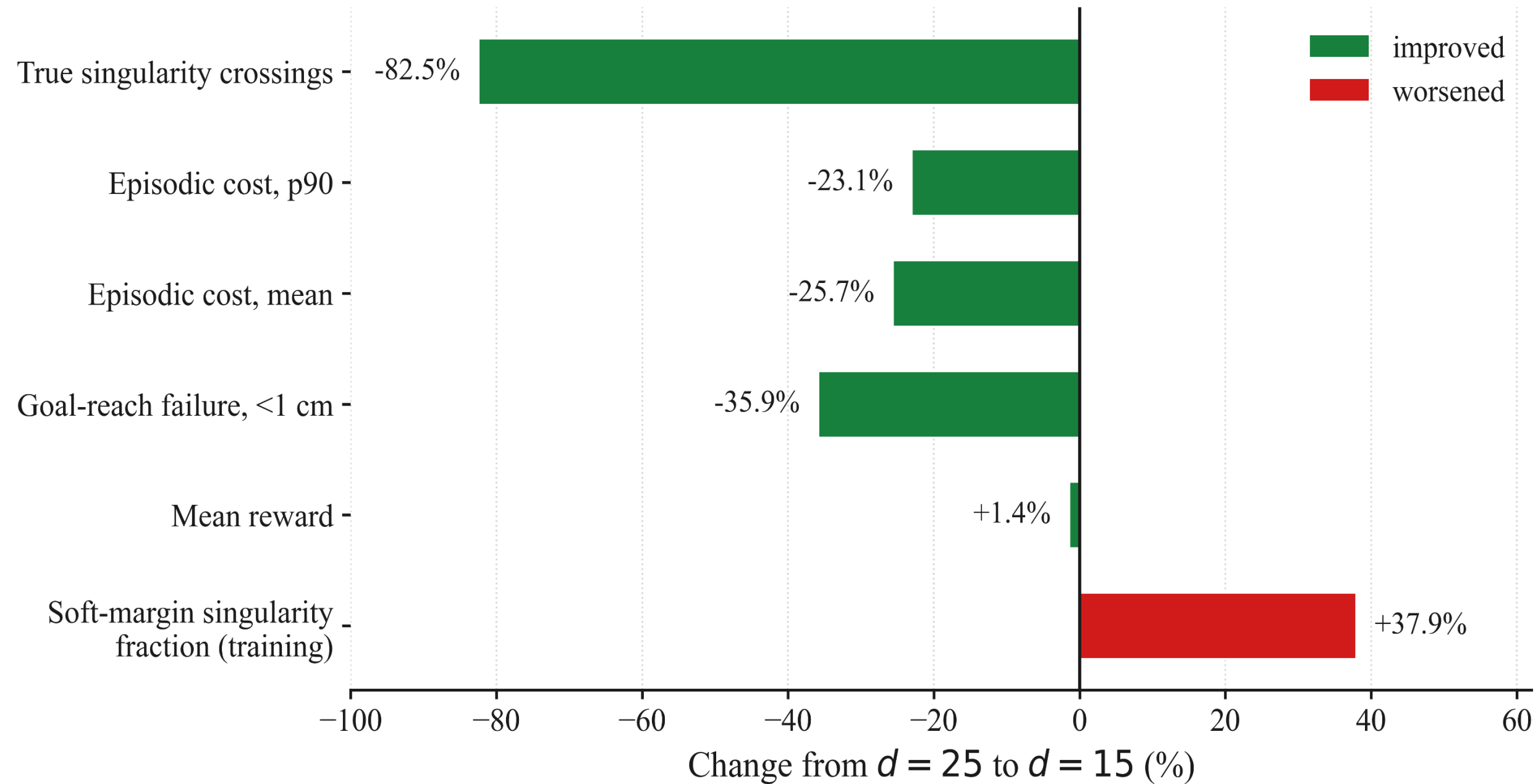


Fig: Effect of tightening the cost limit on the performances

Appendix:

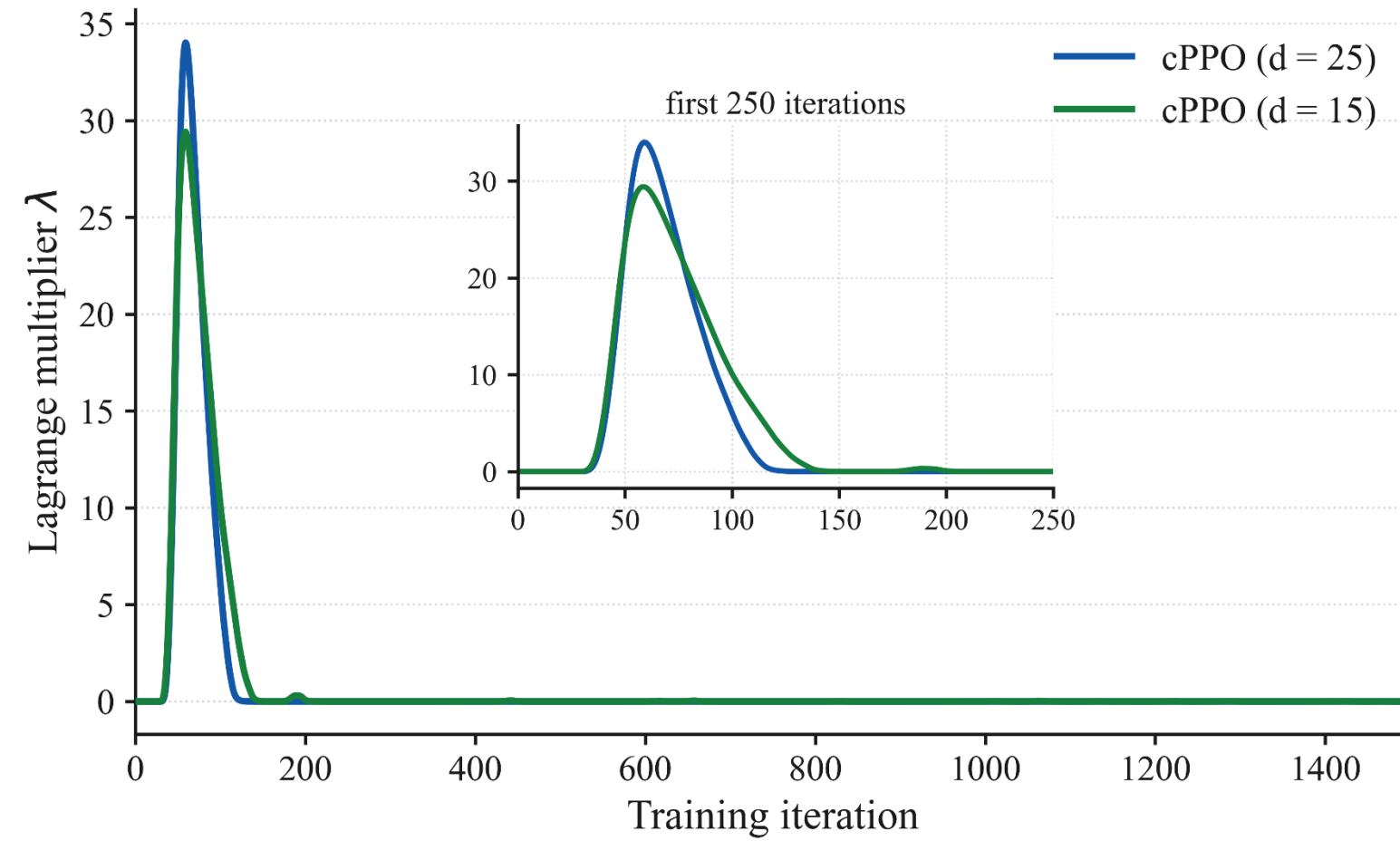


Fig 2: Effect of Lagragian multiplier in action

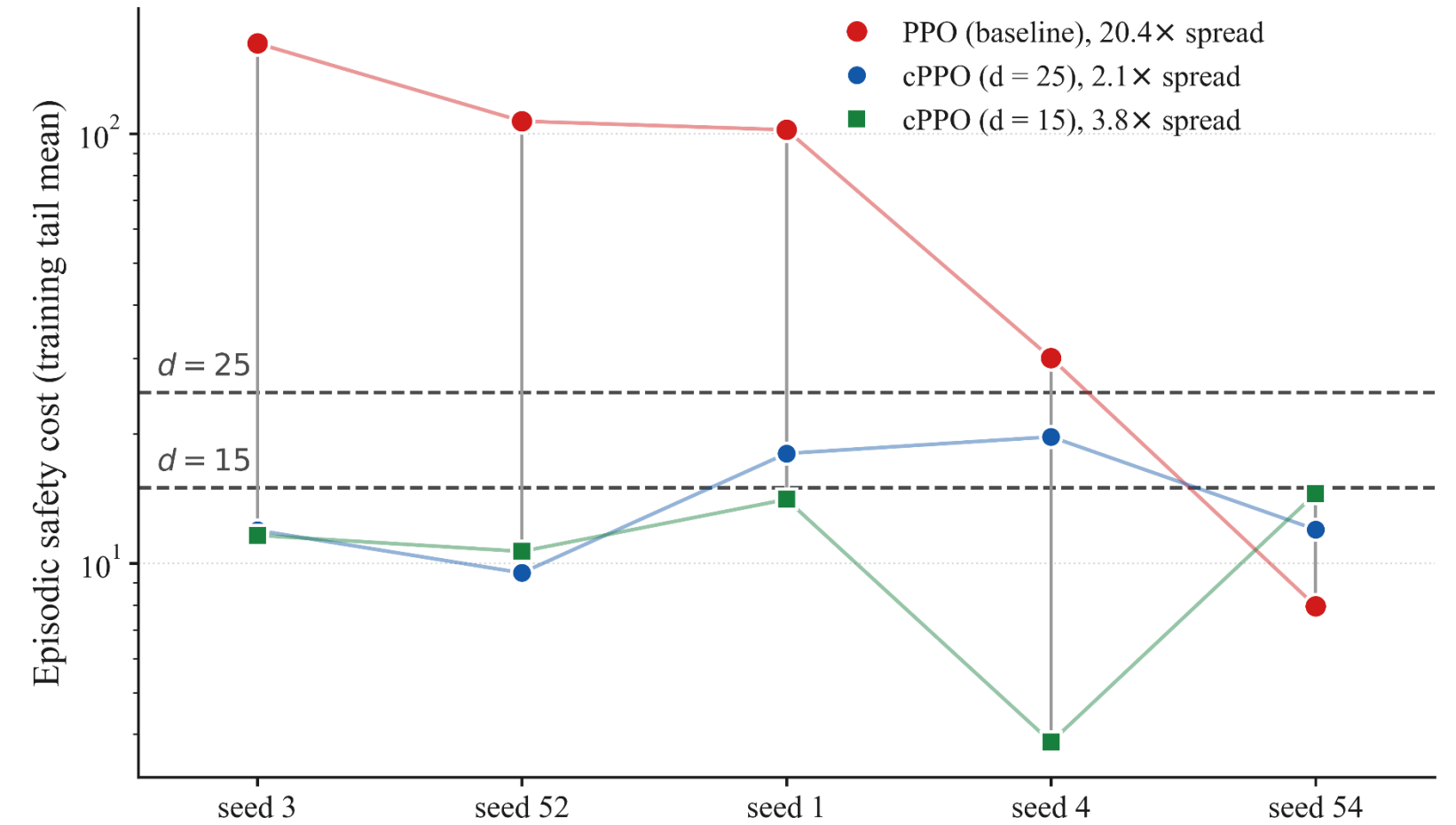


Fig 3: Safety cost varies with the seeds